SOLUTIONS: Control Engineering

1. a) The transfer function for the circuit in Figure 1.2 is given by

$$\frac{Z_f(s)}{Z_i(s)} = -\frac{C_i(s+1/R_iC_i)}{C_f(s+1/R_fC_f)} = -\frac{C_is+1/R_i}{C_fs+1/R_f}$$

Putting in the values: $\boxed{G(s) = G_1(s)G_2(s) = \frac{1}{s(s+2)}}$. [5]

b) Since $y(s) = G(s)u(s)$ we have $\boxed{\ddot{y}(t)+2\dot{y}(t) = u(t)}$. [5]

c) Let $x_1(t) = y(t), x_2(t) = \dot{y}(t)$ and $x(t) = \begin{bmatrix} x_1(t) & x_2(t) \end{bmatrix}^T$. A state–space realisation:

$$\boxed{\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \quad y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)} \qquad [5]$$

d) Since $G(s)$ is type 1, the steady state error to a unit step: $\boxed{e_{ss} = 0.}$. [5]

e) The closed–loop transfer function is $H(s) = \frac{K_p}{s^2+2s+K_p}$. To satisfy the design specifications, the closed–loop poles must be at $-1 \pm j$. Thus $\boxed{K_p = 2.}$ [5]

f) Now, $r(s) = 1/s^2$ and $e(s) = \frac{r(s)}{1+K_pG(s)}$. Therefore:

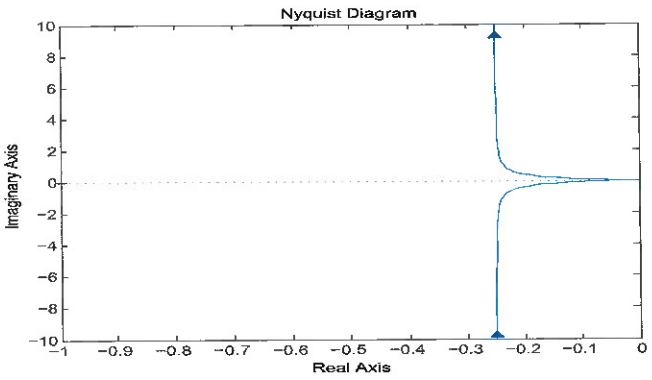
$$\boxed{\lim_{t\to\infty} e(t) = \lim_{s\to 0} se(s) = \lim_{s\to 0} \frac{sr(s)}{1+K_pG(s)} = \lim_{s\to 0} \frac{s+3}{s^2+3s+2} = \frac{3}{2}.} \qquad [5]$$

g) The Nyquist diagram is shown on the next page. [5]

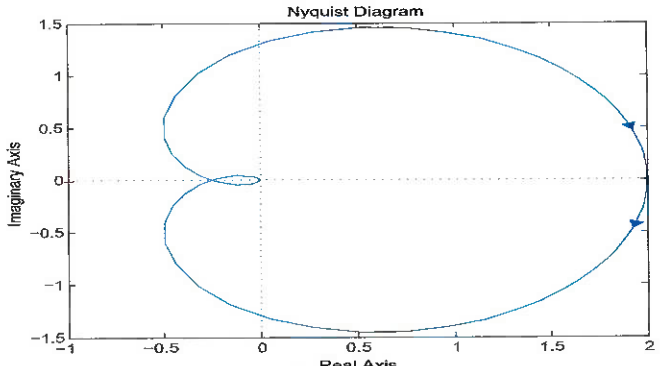
h) The Nyquist criterion: $N = Z - P$ where N is the number of clockwise encirclements of $-1/K_p$, Z is the number of unstable closed loop poles and P is the number of unstable open loop poles ($=0$). So for:

- $\boxed{0 < K_p < \infty \Rightarrow N = 0 \Rightarrow Z = 0}$, $\boxed{-\infty < K_p < 0 \Rightarrow N = 1 \Rightarrow Z = 1}$,
- $\boxed{K_p = 0 \Rightarrow \text{ the closed loop is marginally stable}}$.

[5]


Nyquist Diagram
Imaginary Axis
Real Axis
10
8
6
4
2
0
−2
−4
−6
−8
−10
−1
−0.9
−0.8
−0.7
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0

2. a) The Nyquist diagram is shown below. The real–axis intercepts can be found from the Routh array with $K(s)=K$. The characteristic equation is $1+KG(s)=0$ or $s^3+3s^2+3s+1+2K=0$. The Routh array is



s^3	1	3
s^2	3	$1+2K$
s^1	$(8-2K)/3$	
s^0	$1+2K$	

$K=4$ and $K=-1/2$ give zero row so the real–axis intercepts are $\boxed{-0.25, 2}$ and the corresponding frequencies are $\boxed{\sqrt{3}, 0}$, respectively [6]

b) The closed loop is stable $\boxed{\text{since the first column of the array has positive signs.}}$ The $\boxed{\text{gain margin} = 4}$. At the cross–over frequency ω_c, $|G(j\omega_c)|=1$, or $2=|j\omega_c+1|^3$ or $\boxed{\omega_c \sim 0.766}$ so the $\boxed{\text{phase margin is} \sim 67.6058^\circ}$. [6]

c) A PI compensator has the form $K(s)=K_P+\frac{K_I}{s}=\frac{K_I}{s}(1+\frac{s}{K_I/K_P})$. It has high gain at frequencies below $\omega_0=K_I/K_P$ and gain close to K_P above. The phase is negative and large below ω_0 but insignificant above. It follows that by varying K_I and K_P we can increase low frequency gain (hence improving tracking properties) without introducing phase-lag at high frequency (which would reduce the phase margin) by placing w_0 in the 'middle' frequency range. [6]

d) i) Solving for $\varepsilon(s)$, $\varepsilon(s)=-C(s)\delta(s)$ where $\boxed{C(s)=\frac{G(s)K(s)}{(1+G(s)K(s)}}$. It follows that $\boxed{\text{the loops are equivalent with this value of } C(s)}$. [4]

ii) Since $\Delta(s)$ and the closed loop of Figure 2.1 (and therefore $C(s)$) are stable, the Nyquist stability criterion states that the loop in Figure 2.3 is stable if there are no encirclements by $C(j\omega)\Delta(j\omega)$ of the point -1. The given condition implies that $|C(j\omega)\Delta(j\omega)|<1$ so there are $\boxed{\text{no encirclements by } C(j\omega)\Delta(j\omega) \text{ of } -1 \text{ so the loop is stable}}$. [4]

iii) It follows that the smaller $|C(j\omega)|$ is, the larger the allowed $|\Delta(j\omega)|$ for closed loop stability, and since $|C(j\omega)|=|\frac{K(j\omega)K(j\omega)}{1+G(j\omega)K(j\omega)}|$, and so $\boxed{\text{the smaller the loop gain } |G(j\omega)K(j\omega)| \text{ is, the more robust the closed–loop will be against multiplicative uncertainties}}$.

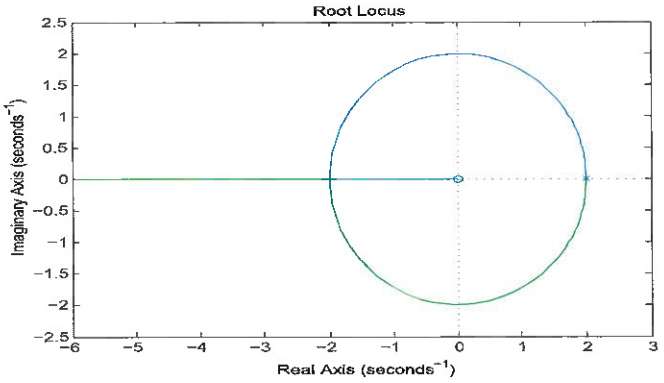
$\boxed{\text{However, reducing the loop gain will result in worse performance}}$ [4]

3. a) For a maximum overshoot of 5% and a settling time of 2 seconds the closed–loop poles must be placed at s_0, $\bar{s}_0 = -2 \pm j2$.

b) For $z=0$, the closed–loop characteristic equation is

$$1+G(s)=0 \Rightarrow 1-\frac{4s}{s^2+ks+4}=0 \Rightarrow s^2-4s+4+ks=0 \Rightarrow 1+k\overbrace{\frac{s}{(s-2)^2}}^{\hat{G}(s)}=0$$

i) We plot the root locus of $\hat{G}$:



ii) Thus a settling time of 2s is only achievable with the closed–loop poles set at -2 and so the response is not oscillatory. The corresponding k is obtained from the gain criterion as $k=-1/\hat{G}(-2)=8$.

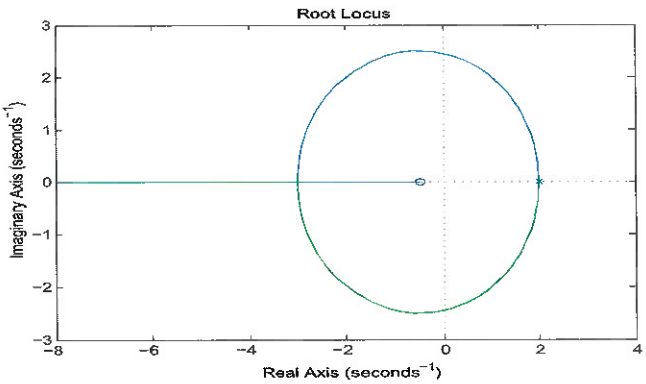
c) For general z, proceeding as before, the closed–loop characteristic equation is

$$1+G(s)=0 \Rightarrow 1-\frac{4s}{s^2+k(s-z)+4}=0 \Rightarrow s^2-4s+4+k(s-z)=0 \Rightarrow 1+k\overbrace{\frac{s-z}{(s-2)^2}}^{\hat{G}(s)}=0$$

i) $\hat{G}(s)$ has two poles at 2 and a zero at z. Let the angle from $s_0=-1+j$ to z be θ and to 2 be θ_1. The angle criterion requires $\theta=2\theta_1-180^\circ$ and after some trigonometry this gives $z=-0.5$.

ii) For $z=-0.5$, the root–locus of $\hat{G}(s)$ is shown below.

iii) The gain criterion requires $k=-1/\hat{G}(s_0)$ and so $k=8$.


Root Locus
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
3
2
1
0
−1
−2
−3
−8
−6
−4
−2
0
2
4